

American Computer Science League

2025 Finals • Short Problems Solutions • Intermediate/Classroom Division

1. Boolean Algebra

$$\begin{aligned} & A(\overline{B} + C) + \overline{B}(C + \overline{A}) + \overline{\overline{B} + C} \\ &= \overline{A}\overline{B} + AC + \overline{B}C + \overline{B}\overline{A} + \overline{B}\overline{C} \\ &= \overline{B}(A + C + \overline{A} + \overline{C}) + AC \\ &= \overline{B} + AC \end{aligned}$$

To be FALSE, $\overline{B} = 0 \rightarrow B = 1 \wedge AC = 0$
Therefore (0,1,0), (0,1,1), and (1,1,0) make it False.

C. (1,1,0) (0,1,*)

2. Boolean Algebra

$$\begin{aligned} & (\overline{A} + AB)(\overline{B}\overline{C}) + (\overline{A} + \overline{BC}) \\ &= \overline{A}\overline{B}\overline{C} + AB\overline{B}\overline{C} + \overline{A} + \overline{B} + \overline{C} \\ &= \overline{B}(\overline{A}\overline{C} + AB\overline{C} + 1) + \overline{A} + \overline{C} \\ &= \overline{B} + \overline{A} + \overline{C} \\ &= \overline{ABC} \end{aligned}$$

D. $\overline{ABC}$

3. Digital Electronics

The Boolean expression for the circuit is:

$$\overline{AB(B + C)} + C$$

C. $\overline{AB(B + C)} + C$

4. Bit-String Flicking

$$\begin{aligned} & (\text{NOT } 000100) \text{ AND } (\text{LSHIFT-3 } (\text{RCIRC-1 } 011011)) \\ &= 111011 \text{ AND } (\text{LSHIFT-3 } 101101) \\ &= 111011 \text{ AND } 101000 \\ &= 101000 \end{aligned}$$

A. 101000

5. Bit-String Flicking

$(10010 \text{ AND } (\text{RSHIFT-1 } 01111)) \text{ OR } X =$
 $(\text{RCIRC-2 } (\text{LSHIFT-2 } 01010 \text{ OR } X))$

Let $X = abcde$.

$\text{LHS} = (10010 \text{ AND } (\text{RSHIFT-1 } 01111)) \text{ OR } X$
 $= (10010 \text{ AND } (\text{RSHIFT-1 } 01111)) \text{ OR } abcde$
 $= 10010 \text{ AND } 00011 \text{ OR } abcde$
 $= 00010 \text{ OR } abcde$
 $= abc1e$

$\text{RHS} = (\text{RCIRC-2 } ((\text{LSHIFT-2 } 01010) \text{ OR } X))$
 $= (\text{RCIRC-2 } ((\text{LSHIFT-2 } 01010) \text{ OR } abcde))$
 $= (\text{RCIRC-2 } (01000 \text{ OR } abcde))$
 $= (\text{RCIRC-2 } a1cde)$
 $= dea1c$

$\text{LHS} = \text{RHS} \rightarrow abc1e = dea1c$

$a = d, b = e, c = a, e = c$

So $a = b = c = d = e$.

Therefore there are 2 solutions: 00000 and 11111

B. 2

6. Recursive Functions

$$f(20) = 2 * f(20 - 3) + 3 = 2 * f(17) + 3 = 2(5) + 3 = 13$$

$$f(17) = 2 * f(17 - 3) + 3 = 2 * f(14) + 3 = 2(1) + 3 = 5$$

$$f(14) = 2 * f(14 - 3) + 3 = 2 * f(11) + 3 = 2(-1) + 3 = 1$$

$$f(11) = f(\lceil 11/2 \rceil) - 2 = f(5) - 2 = 1 - 2 = -1$$

$$f(5) = 5 \bmod 4 = 1$$

$$\text{So } f(f(f(20))) = f(f(13))$$

$$f(13) = 2 * f(13 - 3) + 3 = 2 * f(10) + 3 = 2 * -1 + 3 = 1$$

$$f(10) = f(\lceil 10/2 \rceil) - 2 = f(5) - 2 = 1 - 2 = -1$$

$$\text{So } f(f(f(20))) = f(f(13)) = f(1) = 1 \bmod 4 = 1$$

D. 1

7. Recursive Functions

$$f(20, 5) = f(20 / 2, 5 + 2) + 3 = f(10, 7) + 3 = 19 + 3 = 22$$

$$f(10, 7) = f(10 / 2, 7 + 2) + 3 = f(5, 9) + 3 = 16 + 3 = 19$$

$$f(5, 9) = f((5 - 1) / 2, 9 - 3) - 1 = f(2, 6) - 1 = 17 - 1 = 16$$

$$f(2, 6) = f(2 / 2, 6 + 2) + 3 = f(1, 8) + 3 = 14 + 3 = 17$$

$$f(1, 8) = f((1 - 1) / 2, 8 - 3) - 1 = f(0, 5) - 1 = 15 - 1 = 14$$

$$f(0, 5) = f(0 / 2, 5 + 2) + 3 = f(0, 7) + 3 = 12 + 3 = 15$$

$$f(0, 7) = f(0 / 2, 7 + 2) + 3 = f(0, 9) + 3 = 9 + 3 = 12$$

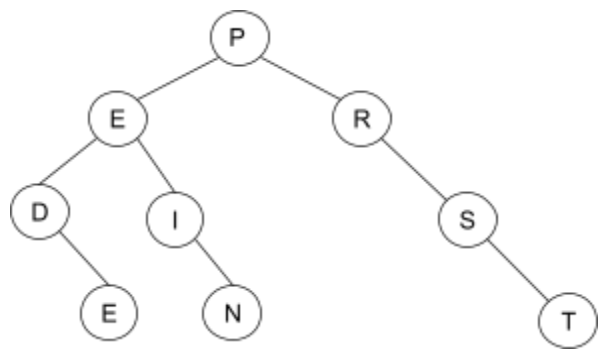
$$f(0, 9) = 0 + 9 = 9$$

D. 22

8. Prefix/Infix/Postfix Notation $\begin{aligned} & - + a / b c * 3 + * a b c \\ & = - + a (/ b c) * 3 + (* a b) c \\ & = - [+a (b c /)] * 3 [+ (a b *) c] \\ & = - (a b c / +) [* 3 (a b * c +)] \\ & = [- (a b c / +) (3 a b * c + *)] \\ & = a b c / + 3 a b * c + * - \end{aligned}$	B. $a b c / + 3 a b * c + * -$																																
9. Prefix/Infix/Postfix Notation $\begin{aligned} & + / * 2 + 6 4 - 9 / 8 2 ^ 4 2 \\ & = + / * 2 (+ 6 4) - 9 (/ 8 2) (^ 4 2) \\ & = + / (* 2 10) (- 9 4) 16 \\ & = + (/ 20 5) 16 \\ & = + 4 16 \\ & = 20 \end{aligned}$	D. 20																																
10. Computer Number Systems The octal numbers from 32_8 to 64_8 : <table><tr><td></td><td>40</td><td>50</td><td>60</td></tr><tr><td></td><td>41</td><td>51</td><td>61</td></tr><tr><td>32</td><td>42</td><td>52</td><td>62</td></tr><tr><td>33</td><td>43</td><td>53</td><td>63</td></tr><tr><td>34</td><td>44</td><td>54</td><td>64</td></tr><tr><td>35</td><td>45</td><td>55</td><td></td></tr><tr><td>36</td><td>46</td><td>56</td><td></td></tr><tr><td>37</td><td>47</td><td>57</td><td></td></tr></table> There are 12 4s in the list.		40	50	60		41	51	61	32	42	52	62	33	43	53	63	34	44	54	64	35	45	55		36	46	56		37	47	57		C. 12
	40	50	60																														
	41	51	61																														
32	42	52	62																														
33	43	53	63																														
34	44	54	64																														
35	45	55																															
36	46	56																															
37	47	57																															
11. Computer Number Systems $\begin{aligned} 3_8 ^ 1 & = 3 \\ 3_8 ^ 2 & = 9_{10} = 11_8 \\ 3_8 ^ 3 & = 27_{10} = 33_8 \\ 3_8 ^ 4 & = 81_{10} = 121_8 \end{aligned}$ Either continue to 7th power or recognize the pattern. If the power is odd, the last octal digit is 3. If the power is even, the last octal digit is 1. Since 7 is odd, the last octal digit is 3.	C. 3																																

12. Data Structures

The binary search tree for **PRESIDENT** is:



To delete node P, the left subtree is moved up and the right child of P is placed on the right branch of N. The number of nodes with only 1 child is 5. They are D, I, N, R, and S.

D. 5

13. Data Structures

The stack is constructed using LIFO as follows:

W → NIL → T → NIL → R → RM → R → RL → R → RN

D. N
R

14. Graph Theory

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix}^2 = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 1 & 1 \end{bmatrix} * \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 & 1 \\ 0 & 2 & 1 & 0 \\ 1 & 0 & 2 & 1 \\ 1 & 2 & 1 & 1 \end{bmatrix}$$

There are 14 paths of length 3.

C. 14

15. Graph Theory

There are 4 cycles in the graph.
They are AEA, BCB, ADEA, and FBCF.

A. 4

16. What Does This Program Do?

```
n = 2025
p = 1
s = 0
while n != 0
    d = n % 10
    s = s + d * p
    p = p * 8
    n = int(n / 10)
end while
output s
```

This program converts 2025_8 to base 10. The variable d gets each digit in reverse order by using $n \% 10$ as n decreases from 2025 to 202 to 20 to 2 to 0. The variable p increases the powers of 8. The variable s gets the sum which is $5*1 + 2*8 + 0*8^2 + 2*8^3 = 1045$.

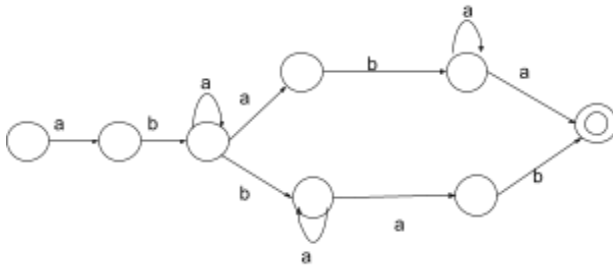
B. 1045

17. LISP

```
(CADAR (REVERSE (CDDR '((a b) c (b (a c)) (a (b) (c))))))
= (CADAR (REVERSE (CDR '(c (b (a c)) (a (b) (c))))))
= (CADAR (REVERSE '((b (a c)) (a (b) (c)))))
= (CADAR '((a (b) (c)) (b (a c))))
= (CADR '(a (b) (c)))
= (CAR '((b) (c)))
= (b)
```

B. (b)

18. FSAs and Regular Expressions



- abaaba - satisfies this FSA
- abbaaba - fails at final a
- ababa - satisfies this FSA
- abbabb - fails at final b
- ababaaa - satisfies this FSA
- abbaaa - missing final b

D. 3

19. FSAs and Regular Expressions

$[\text{^bs}] [\text{aeiou}]^* [\text{bpr}]^* [\text{aeiou}]^* [\text{e-t}]^* \text{s?}$

Each vegetable is checked as follows:

onion - OK

carrot - OK

turnip - OK

peas - OK

asparagus - stops with 5th brace

celery - stops with 5th brace

spinach - stops with 1st brace

beans - stops with 1st brace

broccoli - stops with 1st brace

beet - stops with 1st brace

cauliflower - stops with the 5th brace

cabbage - OK

There are 5 vegetables that satisfy this regular expression. They are:

onion, carrot, turnip, peas, cabbage.

B. 5

20. Assembly Language Programming

T	DC	40	This program tests 41, 42, 43, 44, and 45 to see if $N / T * T = N$ using integer division. Only 45 goes into 2025 evenly. The number 2025 is actually a perfect square since $45^2 = 2025$.
N	DC	2025	
TOP	LOAD	T	
	ADD	=1	
	STORE	T	
	LOAD	N	
	DIV	T	
	MULT	T	
	SUB	N	
	BE	STOP	
	BU	TOP	
STOP	PRINT	T	
	END		

C. 45